

HR PAYSリップ AUTOMATION

Updated Implementation, Architecture, Calculation & Control Research (Post-Review)

0. Executive Summary

This is the updated, evidence-based implementation blueprint for the InterTech Systems Payslip Automation module. It was produced against four inputs, per the required source-priority order: (1) the current Billing Rate workbook actually inspected — "Current_BILLING RATE_ANALYSIS_ODUM 19TH AUGUST 2026_18TH SEPTEMBER 2026 (BASE FEE + ALLOWANCE).xlsx" — and the current Payslip template structure evidenced in "FEBRUARY 2026 PAYSLIP -Seun Odufejo.xlsx"; (2) HR's written answers to the v1 report's Risks and Unresolved Questions; (3) the v1 report's own findings; and (4) fresh architectural recommendations, with assumptions labelled as such throughout.

Inspecting the current workbook directly, rather than relying on the v1 mapping was essential: the ALLOWANCE SPREAD SHEET and INVOICE SPREAD SHEET column layouts have materially changed since the January/February 2026 workbook, and several v1 recommendations (including its own best-guess column letters for Performance Bonus, 13th Month, Christmas Bonus, and Marriage Support Grant) turn out to be wrong against the current file. This is not a criticism of v1's method, it correctly labelled those guesses as unverified. It is the concrete proof of why this module must never hard-code column letters, and it is treated as the single most important finding carried into this update.

Five findings materially change or sharpen the v1 design:

- **CURRENT ARTIFACT EVIDENCE:** the ALLOWANCE SPREAD SHEET has grown from 48 to 59 columns and almost every taxable-bonus column letter has shifted (e.g. Performance Bonus is AO, not the AM guessed in the research prompt; 13th Month is AR, not AP; Christmas Bonus is AU, not AS; Marriage Support Grant is AX, not AV). Nine allowances now have an explicit adjacent "ARREARS —" companion column already built into the template.
- **CURRENT ARTIFACT EVIDENCE:** the updated payslip template confirms HR's answer precisely — its live Net Service Fee formula is " $=C41-E27+E30+E31+E32+E33+E34+E35-E38$ ", i.e. Gross Earnings – Total Deductions + Transport + Accommodation + Feeding + OutStation + Christmas Bonus + Marriage Support Grant – Other Deduction. Gross Earnings is " $=SUM(C26:C40)$ ", which includes two deliberately blank buffer rows (39–40) — the template's own low-tech mechanism for future line items without a formula change.
- **CURRENT ARTIFACT EVIDENCE:** the INVOICE SPREAD SHEET now carries a per-employee "LOAN / ADVANCE – DEDUCTIONS" column (M). HR's answer states Other Deductions remain a manual, post-reconciliation HR entry — but this column's existence is a genuine new contradiction between the current artifact and the current business answer, not something to resolve silently (Section 26).
- **ROW-VALIDITY CORRECTION:** the current Odum workbook has 599 numbered rows, but only 76 carry a real employee (Staff ID + Personnel + Base Fee all present); rows 81–599 are blank template rows pre-numbered for future joiners, and row 600 is a TOTAL summary row. An importer that counted any row with a non-blank cell would have overstated this company's headcount roughly eightfold — the three-field "real employee row" test in §8.1 is now a mandatory, load-bearing parsing rule, not a nicety. With this correction, Odum's 76 employees sit comfortably within HR's ~570-across-four-companies estimate (Section 21), which is no longer treated as a discrepancy.
- The Billing Rate now embeds accounting-company identity directly (Input tab: Accounting Company, Paying Company, Period, Month, Year, Commission %, VAT %; repeated per employee row) — confirming HR's answer that each accounting company maintains and uploads its own workbook, and giving the multi-company design (Section 13/21) a concrete, evidenced anchor rather than an assumption.

The recommended architecture is unchanged in its outer shape — Import → Validate → Normalize → Calculate → Generate → Review/Correct → Approve → Deliver → Audit, with immutable approved versions — but is now extended with: a header-driven (not position-driven) mapping layer proven necessary by the observed column drift; an explicit allowance-classification configuration table instead of a hard-coded list; a dynamic arrears model that mirrors the workbook's own "allowance + adjacent arrears column" pattern; an HR Excel review/correction loop with protected, non-editable

calculated cells; a multi-company/payroll-period data model; and a payslip-generation decision revisited in light of the now-required dynamic line items.

1. Scope Boundary

Unchanged from v1 and re-confirmed by the current artifacts. Upstream (Proforma → Attendance → Civil Time Sheet → Billing Rate) remains outside this module and outside HR’s Payslip Automation ask.

In scope	Out of scope
Billing Rate upload, validation, normalization	Proforma, Attendance Sheet, Civil Time Sheet preparation
Deterministic calculation (Consultant Gross Pay → Gross Earnings → WHT → Net Service Fee)	Billing Rate preparation by the four accounting companies
Payslip generation, HR review/correction, approval, versioning	Full statutory payroll (pension, NHF, remittances) and its filing
HR-triggered email delivery; employee-requested WhatsApp delivery	Automated Payroll/bank-instruction reconciliation — remains deliberately manual (HR answer, Section 5.9 legacy / §12 this doc)
Audit logging, versioning, multi-company/period tracking	—

2. Current Source Artifact Analysis

2.1 Current Billing Rate workbook

FACT (observed in current artifacts): the workbook "Current_BILLING RATE_ANALYSIS_ODUM 19TH AUGUST 2026_18TH SEPTEMBER 2026 (BASE FEE + ALLOWANCE).xlsx" has five tabs — Dropdown, Input, INVOICE SPREAD SHEET, ALLOWANCE SPREAD SHEET, Invoice Page Template — identical tab names to the January/February 2026 workbook used in v1. ALLOWANCE SPREAD SHEET now spans A:BG (59 columns, up from 48) with 599 numbered rows; INVOICE SPREAD SHEET spans A:Z (up from A:J) and carries the same row structure. Of those 599 rows, only 76 (rows 5–80) carry a real employee — Staff ID, Personnel, and Base Fee all populated. Rows 81–599 are pre-numbered (S/N filled) but otherwise blank template rows, evidently left ready for new joiners to be added without restructuring the sheet; row 600 is a TOTAL summary row (a formula, not an employee). Section 8.1 makes the three-field completeness test the formal rule for what counts as an employee row.

Input tab field	Observed value (this upload)
Accounting Company	ODUM ENERGY FZE - TRAIN A
Paying Company	INTERTECH SYSTEMS LIMITED
Period	19-AUGUST-2026 TO 18-SEPTEMBER-2026
Month / Year	SEPTEMBER / 2026
Commission % / VAT %	2% / 0% (client-invoice inputs — not used by the payslip calculation, Section 2.3)

2.2 Current INVOICE SPREAD SHEET header row (row 4)

Column	Header observed	Role
A	S/N	Row sequence, not an identity field
B	STAFF ID	Primary employee key (unchanged position from v1)

Column	Header observed	Role
C	PERSONNEL	Display/audit only
D	BASE FEES	baseFee source (unchanged position)
E	JOB TITLE	New — display/audit only
F	ACCOUNTING COMPANY	New — formula "=\$D\$1" pulling the Input-tab value onto every row
G	PAYING COMPANY	New — formula "=\$D\$2"
H	DAYS IN THE MONTH	totalDays source (unchanged position)
I	DAYS ABSENT	New, explicit
J	DAYS WORKED	Now a FORMULA, "=H-I" (Total Days minus Days Absent) — no longer a direct HR entry
K	SERVICE FEES	Formula "=D/H*J" — this is Consultant Gross Pay, already computed in-sheet
L	ABSENT	Formula "=D-K" — the fee value attributable to absence, informational
M	LOAN / ADVANCE - DEDUCTIONS	New, per-employee, currently 0 on inspected rows — see §26 for the contradiction this raises against HR's "manual Other Deduction" answer
N, O	REIMBURSABLE / NON-REIMBURSABLE ALLOWANCES TOTAL	Pull the ALLOWANCE SPREAD SHEET's BD/BE totals — client-invoice inputs, not payslip inputs (Section 2.3)
P-T	Invoice subtotal, 1% fee, commission, VAT, invoice total	Feed the agency's client invoice (Invoice Page Template) — out of scope for the payslip engine

2.3 A distinction the workbook makes that the payslip design must respect

The INVOICE SPREAD SHEET computes two different things that must never be conflated: (a) columns B–M are per-employee payroll inputs the payslip engine needs, and (b) columns N–T compute what InterTech invoices its client for that employee's services (reimbursable/non-reimbursable allowance totals, a 1% fee, commission, VAT). The client-invoice figures are structurally unrelated to the employee's Net Service Fee and must not be imported into the calculation engine's domain model at all — doing so would be a category error, not just an unused field.

2.4 Current ALLOWANCE SPREAD SHEET header row (row 4) — all 59 columns

Verified column-by-column against the file. Columns not referenced elsewhere in this document are Billing Rate fields the payslip does not currently use (Section 5).

Cols	Headers observed (in column order)
A–F	S/N · STAFF ID · PERSONNEL · BASE FEE · ACCOUNTING COMPANY · PAYING COMPANY
G–J	ACCOMMODATION ALLOWANCE · ARREARS ACCOMMODATION ALLOWANCE · TIME OFF TRANSPORT ALLOWANCE · ARREARS TIME OFF TRANSPORT ALLOWANCE
K–N	TRANSPORT ALLOWANCE · ARREARS TRANSPORT ALLOWANCE · FEEDING/MISSED MEAL ALLOWANCE · ARREARS FEEDING/MISSED MEAL ALLOWANCE
O–R	OUT STATION DUTY ALLOWANCE · ARREARS OUT STATION DUTY ALLOWANCE · HOTEL ACCOMMODATION ALLOWANCE · RECHARGE CARD ALLOWANCE
S–U	CERTIFICATE / COURSE REIMBURSEMENT · MOBILIZATION ALLOWANCE · RE-LOCATION ALLOWANCE
V–Y	PUBLIC HOLIDAY ALLOWANCE · ARREARS PUBLIC HOLIDAY ALLOWANCE · WEEKEND WORK ALLOWANCE · ARREARS WEEKEND WORK ALLOWANCE

Cols	Headers observed (in column order)
Z-AB	HOURLY/WEEKDAY OVERTIME ALLOWANCE · ARREARS HOURLY/WEEKDAY OVERTIME ALLOWANCE · ARREARS SERVICE FEES
AC-AD	PRESENT ON TIME OFF (= Extra Day Work source, HR-confirmed) · ARREARS PRESENT ON TIME OFF
AE-AN	FIELD · SHIFT · RIG · JOB BONUS · KRA · HAZARD · HSC BONUS · MARINE · R & M BONUS · LAB BONUS (allowances)
AO-AR	PERFORMANCE BONUS · END OF CONTRACT BONUS · JOINING BONUS · 13TH MONTH BONUS
AS-AV	ANNUAL LEAVE ALLOWANCE · PRODUCTION BONUS · CHRISTMAS BONUS · LEAVE ENCASHMENT ALLOWANCE
AW-BC	NOTICE PAY · MARRIAGE SUPPORT GRANT · LONG SERVICE AWARD · LOYALTY BONUS · TERMINAL BENEFIT · MEDICAL ALLOWANCE · SPECIAL MGT APPROVED BONUS
BD-BG	REIMBURSABLE ALLOWANCES TOTAL · NON-REIMBURSABLE ALLOWANCES TOTAL · TOTAL ALLOWANCE · REMARKS

2.5 Current payslip template (structure evidenced via "FEBRUARY 2026 PAYSリップ -Seun Odufejo.xlsx")

This file is used here purely as structural evidence of the current, updated payslip template — its specific figures belong to a different period than the Odum workbook above and are not mixed into the worked example in Section 7.

Row range	Left column (earnings, C)	Right column (deductions/allowances, E)
23–26	Base Fee · Days Worked · Total Days · Consultant Gross Pay ("=C23*(Worked/Total)")	WHT ("=0.05*C41")
27–38	End of Contract Bonus · Joining Bonus · Arrears Service Fees · Leave Encashment · Overtime · Rig Bonus · Job Bonus · Weekends · Extra Day Work · Public Holiday · Performance Bonus · 13th Month	Total Deductions ("=E23") · Transport · Accommodation · Feeding · OutStation · Christmas Bonus · Marriage Support Grant · (blank) · (blank) · Deduction ("Other Deduction")
39–40	Blank — reserved buffer rows, included in the SUM range	—
41	Gross Earnings = "=SUM(C26:C40)"	Net Service Fee = "=C41-E27+E30+E31+E32+E33+E34+E35-E38"

3. Former Research vs. Updated HR Decisions — Risk / Resolution Table

This is the mandatory Risk/Resolution table, consolidating v1's numbered Risks & Unresolved Questions, HR's written answers, and this update's independent verification against the current artifacts. It doubles as the response to Section 26 of the research brief — no item below is re-opened without new artifact evidence.

Previous item	HR answer	Current status	Implementation impact	Remaining risk
Net Service Fee: 3 vs 4 allowance terms in the old payslip formula	Oversight in the old manual payslip; standardize on 4 non-taxable allowances	RESOLVED + exceeded: current template adds Christmas Bonus and Marriage Support Grant as two more add-terms (6 total)	Net Service Fee formula updated to Gross Earnings – Total Deductions + Transport + Accommodation + Feeding + OutStation + Christmas Bonus + Marriage Support Grant – Other Deduction	Low — directly observed in the current template
Days Worked/Total Days mismatch (Billing Rate vs. old payslip)	Oversight; Billing Rate is always the source of truth	RESOLVED, and reinforced: Days Worked is now a formula in the Billing Rate itself (Total Days – Days Absent), removing the chance of an independently typed, conflicting value	Normalization layer takes Days Worked only from the Billing Rate's computed value; never accepts a payslip-side days figure	Low
Other Deduction source unconfirmed	Manual HR entry, post-payroll-reconciliation, deliberately human-only	PARTIALLY RE-OPENED BY NEW EVIDENCE: the current workbook now has a per-employee "LOAN / ADVANCE – DEDUCTIONS" column not present in the older file	MVP still treats Other Deduction as HR-entered per Section 5.3/6.4 of the brief, per the explicit HR answer	The new Loan/Advance column isn't meant to seed Other Deduction automatically, as it is unrelated/for a different purpose
ExtraDayWork source unconfirmed	Maps to PRESENT ON TIME OFF	RESOLVED and verified: column AC on the current ALLOWANCE SPREAD SHEET, with its own arrears companion (AD)	extraDayWork normalized field sources from AC; treated as taxable per its position among the payslip's taxable rows (row 35)	Low
Unmapped allowance columns (Field, Shift, KRA, Hazard, etc.)	Ignore — not currently paid; only payslip-mapped items matter	RESOLVED as a classification rule, not a one-off list: any column not on the current payslip is IGNORED by default (Section 5)	Allowance classification becomes configuration-driven so newly added workbook columns default to IGNORED until an HR-approved reclassification, rather than needing another one-off list	Low, provided the classification table is kept current when the workbook template changes
Performance Bonus / 13th Month tax treatment	Taxable	RESOLVED; source columns verified as AO (Performance Bonus) and AR (13th	Header-driven mapping, not position guessing, prevents this class of error going forward	Low, now that both are directly observed

Previous item	HR answer	Current status	Implementation impact	Remaining risk
		Month) — both DIFFERENT from the AM/AP guessed in the research brief		
Christmas Bonus / Marriage Support Grant tax treatment	Non-taxable	RESOLVED; source columns verified as AU (Christmas Bonus) and AX (Marriage Support Grant) — both different from the AS/AV guessed	Both added to the four standard non-taxable allowances as payslip-mapped non-taxable fields	Low
Dynamic allowance/arrears requirement	Wanted so arrears can sit beneath an allowance and be summed into Gross Earnings/Net Service Fee	PARTIALLY EVIDENCED: the Billing Rate already has 9 built-in "ARREARS —" companion columns; the payslip template's dynamism is currently just 2 blank buffer rows, not true dynamic rows	Section 10 designs a domain-level dynamic-line model rather than relying on the Billing Rate's fixed arrears columns or the payslip's 2 spare rows, since neither structurally scales to "any allowance, any number of arrears"	Medium — requires moving payslip generation off a fixed-row Excel range (Section 11)
Payroll reconciliation automation	Deliberately manual, human-only	CONFIRMED, unchanged	No automated name-matching; a reconciliation view only (Section 12)	Low, by design
WhatsApp identity verification	Request Staff ID before sending	CONFIRMED; adopted as the second factor after phone-number identification	Section 17 details rate limiting and audit controls layered around this factor, since Staff ID alone is a modest control for compensation data	Medium — Staff IDs are not secret to co-workers; see Section 17 for compensating controls
Workforce scale	~570 employees, 4 accounting companies, each uploading its own Billing Rate	CONFIRMED, once corrected for template noise: Odum's workbook has 599 numbered rows but only 76 real employees (Staff ID + Personnel + Base Fee populated); the rest are blank rows reserved for future joiners plus one TOTAL row. 76 per company is consistent with a ~570 total across four companies	Section 21 sizes the architecture to HR's ~570 figure; Section 8.1 makes the three-field row-validity test mandatory so no future importer miscounts template rows as employees	Low — resolved by re-reading the file with the correct row filter, not by a new HR answer

4. Billing Rate Workbook Mapping — Source Mapping Table

This is the mandatory Source Mapping Table. It supersedes v1's mapping in full; every row is verified against the current workbook, not carried over from the old one.

Domain field	Source sheet	Source header	Col	Tax class	In Gross Earnings?	In Net Service Fee?	Arrears?
staffId	INVOICE SPREAD SHEET	STAFF ID	B	n/a	n/a	n/a	n/a
baseFee	INVOICE SPREAD SHEET	BASE FEES	D	n/a	input	input	No
totalDays	INVOICE SPREAD SHEET	DAYS IN THE MONTH	H	n/a	input	input	No
daysAbsent	INVOICE SPREAD SHEET	DAYS ABSENT	I	n/a	input	input	No
daysWorked	INVOICE SPREAD SHEET	DAYS WORKED (formula = H-I)	J	n/a	input	input	No
accountingCompany	INVOICE SPREAD SHEET / Input	ACCOUNTING COMPANY	F (row) / B3 (Input)	n/a	no	no	No
consultantGrossPay	Derived	baseFee × (daysWorked / totalDays)	—	Taxable	Yes	Yes	No
endOfContractBonus	ALLOWANCE SPREAD SHEET	END OF CONTRACT BONUS ALLOWANCE	AP	Taxable	Yes	via Gross Earnings	No
joiningBonus	ALLOWANCE SPREAD SHEET	JOINING BONUS ALLOWANCE	AQ	Taxable	Yes	via Gross Earnings	No
arrearsServiceFees	ALLOWANCE SPREAD SHEET	ARREARS SERVICE FEES	AB	Taxable	Yes	via Gross Earnings	No (it is itself an arrears line)
leaveEncashment	ALLOWANCE SPREAD SHEET	LEAVE ENCASHMENT ALLOWANCE	AV	Taxable	Yes	via Gross Earnings	No
overtime	ALLOWANCE SPREAD SHEET	HOURLY/WEEKDAY OVERTIME ALLOWANCE	Z	Taxable	Yes	via Gross Earnings	Yes (AA)
rigBonus	ALLOWANCE SPREAD SHEET	RIG ALLOWANCE	AG	Taxable	Yes	via Gross Earnings	No

Domain field	Source sheet	Source header	Col	Tax class	In Gross Earnings?	In Net Service Fee?	Arrears?
jobBonus	ALLOWANCE SPREAD SHEET	JOB BONUS ALLOWANCE	AH	Taxable	Yes	via Gross Earnings	No
weekends	ALLOWANCE SPREAD SHEET	WEEKEND WORK ALLOWANCE	X	Taxable	Yes	via Gross Earnings	Yes (Y)
extraDayWork	ALLOWANCE SPREAD SHEET	PRESENT ON TIME OFF	AC	Taxable	Yes	via Gross Earnings	Yes (AD)
publicHoliday	ALLOWANCE SPREAD SHEET	PUBLIC HOLIDAY ALLOWANCE	V	Taxable	Yes	via Gross Earnings	Yes (W)
performanceBonus	ALLOWANCE SPREAD SHEET	PERFORMANCE BONUS	AO	Taxable	Yes	via Gross Earnings	No
thirteenthMonth	ALLOWANCE SPREAD SHEET	13TH MONTH BONUS ALLOWANCE	AR	Taxable	Yes	via Gross Earnings	No
transportAllowance	ALLOWANCE SPREAD SHEET	TRANSPORT ALLOWANCE	K	Non-taxable	No	Yes (direct add)	Yes (L)
accommodationAllowance	ALLOWANCE SPREAD SHEET	ACCOMMODATION ALLOWANCE	G	Non-taxable	No	Yes (direct add)	Yes (H)
feedingAllowance	ALLOWANCE SPREAD SHEET	FEEDING/MISSED MEAL ALLOWANCE	M	Non-taxable	No	Yes (direct add)	Yes (N)
outStationAllowance	ALLOWANCE SPREAD SHEET	OUT STATION DUTY ALLOWANCE	O	Non-taxable	No	Yes (direct add)	Yes (P)
christmasBonus	ALLOWANCE SPREAD SHEET	CHRISTMAS BONUS ALLOWANCE	AU	Non-taxable	No	Yes (direct add)	No
marriageSupportGrant	ALLOWANCE SPREAD SHEET	MARRIAGE SUPPORT GRANT	AX	Non-taxable	No	Yes (direct add)	No
otherDeduction	Manual HR input	(none — see §26 re: LOAN/ADVANCE column M)	—	Deduction	No	Yes (subtract, via Total Deductions)	No
wht	Derived	5% × Gross Earnings	—	Deduction	No	Yes (subtract, via Total Deductions)	No

5. Allowance Classification Matrix

SUPPLIED BUSINESS RULE: every Billing Rate allowance that does not appear on the current payslip is IGNORED because it is not currently paid to employees. This section makes that an explicit, queryable classification rather than an implicit omission, so a future workbook column default to IGNORED until an HR-approved reclassification — never to "calculated by accident."

Source allowance (Billing Rate)	Classification	Reason	Payslip label	Tax treatment
Accommodation, Time Off Transport, Transport, Feeding, OutStation (+ their arrears)	PAYSLIP-MAPPED	Appear on the current payslip	Accommodation / Transport / Feeding / OutStation Allowance	Non-taxable
Public Holiday, Weekend Work, Overtime, Present On Time Off, Arrears Service Fees (+ their arrears)	PAYSLIP-MAPPED	Appear on the current payslip	Public Holiday / Weekends / Overtime worked / Extra Day Work / Arrears service fees	Taxable
End of Contract Bonus, Joining Bonus, Leave Encashment, Rig, Job Bonus, Performance Bonus, 13th Month	PAYSLIP-MAPPED	Appear on the current payslip	(same names)	Taxable
Christmas Bonus, Marriage Support Grant	PAYSLIP-MAPPED	Appear on the current payslip (added since v1)	(same names)	Non-taxable
Hotel Accommodation, Recharge Card, Certificate/Course Reimbursement, Mobilization, Re-location	IGNORED / NOT CURRENTLY PAID	Not present on the current payslip; HR confirmed unused columns are unpaid	—	n/a
Field, Shift, KRA, Hazard, HSC Bonus, Marine, R&M Bonus, Lab Bonus	IGNORED / NOT CURRENTLY PAID	Same as above	—	n/a
Annual Leave, Production Bonus, Notice Pay, Long Service Award, Loyalty Bonus, Terminal Benefit, Medical Allowance, Special Mgt Approved Bonus	IGNORED / NOT CURRENTLY PAID	Same as above	—	n/a
Loan / Advance – Deductions (INVOICE SPREAD SHEET, column M)	REQUIRED BUT UNMAPPED	New column; conflicts with HR's "manual Other Deduction" answer — needs one confirmation before use (§26/27)	"Deduction" (currently HR-typed)	Deduction
Reimbursable / Non-Reimbursable / Total Allowance totals (BD/BE/BF)	IGNORED FOR PAYSLIP (used only for the client invoice, §2.3)	Feed the Invoice Page Template, not the payslip	—	n/a

5.1 Configuration model recommendation

RECOMMENDATION: back this table with a database-backed, versioned Allowance Definition table rather than a hard-coded enum or a spreadsheet of rules. Each row: canonical field name, source sheet, source header (the alias actually matched, for audit), classification (PAYSLIP-MAPPED / IGNORED / MANUAL / UNMAPPED), tax category, payslip display label and position, whether it affects Gross Earnings, whether it affects Net Service Fee directly, whether arrears are supported for it, and an effective-from date. This is a configuration table, not application code, specifically because the current artifact evidence (Section 3) shows this list already changed once between the two research passes and should be changeable by an authorized HR/engineering pair without a deployment.

6. Calculation Specification

6.1 Consultant Gross Pay

$\text{consultantGrossPay} = \text{baseFee} \times (\text{daysWorked} / \text{totalDays})$
daysWorked is itself derived in the Billing Rate as $\text{totalDays} - \text{daysAbsent}$ —
the normalization layer takes the Billing Rate's own computed value, never re-derives it independently.
Guard: $\text{totalDays} = 0$ is a blocking validation error (never divide).

6.2 Gross Earnings

$\text{grossEarnings} = \text{consultantGrossPay}$
+ SUM(all fields classified PAYSLIP-MAPPED · Taxable)
+ SUM(any approved dynamic arrears lines attached to a taxable allowance, \$10)
// Never hard-code an Excel range (e.g. "SUM(C26:C40)"); the payslip's own SUM range is presentation
// only — the engine sums named, classified components from the AllowanceDefinition table (§5.1).

6.3 Withholding Tax (WHT)

$\text{wht} = 0.05 \times \text{grossEarnings}$
SUPPLIED BUSINESS RULE: applies once, to every contract-staff employee, and only through Total Deductions — never subtracted a second time elsewhere.

6.4 Total Deductions

$\text{totalDeductions} = \text{wht} + \text{otherDeduction}$
otherDeduction: HR-entered after manual payroll reconciliation (default 0).
otherDeduction is NOT sourced from the Billing Rate's LOAN/ADVANCE column in v1 — see §3/26.

6.5 Net Service Fee

$\text{netServiceFee} = \text{grossEarnings}$
- totalDeductions
+ transportAllowance + accommodationAllowance + feedingAllowance + outStationAllowance
+ christmasBonus + marriageSupportGrant
+ SUM(any approved dynamic arrears lines attached to a non-taxable allowance, \$10)

FACT (observed in current artifacts): this six-add-term formula is read directly off the current payslip's live formula ("=C41-E27+E30+E31+E32+E33+E34+E35-E38"), not inferred — it is the strongest possible evidence for the calculation engine's Net Service Fee logic.

6.6 Stability check (per the brief's explicit requirement)

Scenario	Formula behaviour
One or more non-taxable allowances = 0	Adds 0; no special-case needed — confirmed by the current inspected workbook, where most non-taxable fields are blank for most employees
All non-taxable allowances populated	All six add-terms apply; no interaction effects between them
Allowance arrears exist	Arrears amounts are added as their own line items (§10), inheriting the parent allowance's tax class, before the Gross Earnings/Net Service Fee sums run — they are never merged into the base allowance figure, so the calculation trace shows both amounts separately
Other Deductions exist	Subtracted once, inside totalDeductions, alongside WHT
A taxable bonus exists (e.g. Performance Bonus)	Included in grossEarnings, which flows into both WHT (increasing it) and Net Service Fee (as part of Gross Earnings)
A non-taxable bonus exists (e.g. Christmas Bonus)	Excluded from grossEarnings/WHT; added directly to Net Service Fee only

7. Calculation Example

FACT (observed in current artifacts): worked from row 16 of the current, actually-inspected Odum workbook — Staff ID 2230355, Uche Emmanuel Chukwuma, Mechanical Technician, period 19 Aug–18 Sep 2026 — chosen because it is one of the few rows in this upload carrying a non-zero payslip-mapped allowance, giving a non-trivial example.

Field	Source	Value
Base Fee	INVOICE SPREAD SHEET D16	₦ 776,160
Total Days / Days Absent / Days Worked	H16 / I16 / J16 (=H-I)	31 / 0 / 31
Consultant Gross Pay	$776,160 \times (31/31)$	₦ 776,160
Public Holiday (taxable)	ALLOWANCE SPREAD SHEET V16	₦ 51,744
All other payslip-mapped allowances	Blank on this row	₦ 0
Gross Earnings	$776,160 + 51,744$	₦ 827,904
WHT	$5\% \times 827,904$	₦ 41,395.20
Other Deduction	Not yet reconciled for this period	₦ 0
Total Deductions	$41,395.20 + 0$	₦ 41,395.20
Non-taxable allowances (Transport/Accommodation/Feeding/OutStation/Christmas/Marriage)	All blank on this row	₦ 0
Net Service Fee	$827,904 - 41,395.20 + 0$	₦ 786,508.80

8. Excel Ingestion Architecture

v1 recommended reusing the existing SheetJS/xlsx-based Employee Import pattern for reading. That recommendation is confirmed for reading, with one refinement forced by the column-drift evidence in Section 2/3: the importer must resolve fields by header text, never by column position, with position recorded only as an audit fact.

8.1 Header-driven, versioned mapping layer

1. Load the current AllowanceDefinition/mapping configuration (versioned, Section 5.1) for the workbook's detected structure.
2. Row-validity filter (mandatory): a row is treated as a real employee only if Staff ID, Personnel, and Base Fee are ALL present and non-zero. The current Odum workbook proves why this cannot be a soft heuristic: it carries 599 numbered rows but only 76 real employees, with rows 81–599 pre-numbered (S/N filled) and otherwise blank, evidently reserved so HR can add a new joiner without restructuring the sheet, and row 600 is a TOTAL formula row. Any row failing this three-field test is silently skipped — not validated, not reported as an error — since it is template scaffolding, not data. A row with exactly one or two of the three fields populated (a genuine partial entry) is different and IS reported as a validation error (Section 16), since that pattern cannot be template noise.
3. Confirm the expected sheet names exist (Input, INVOICE SPREAD SHEET, ALLOWANCE SPREAD SHEET); the other two tabs are not read.
4. For each expected field, scan the header row (row 4 on both spread sheets) for its configured header text or an approved alias; record the column actually matched.
5. If any required header is missing, or an approved header text now resolves to a different column than the last successful import, treat this as structural drift: fail the whole upload with a message naming the sheet, the expected header, and what (if anything) was found — do not guess or shift.
6. A brand-new, unrecognized header column is neither an error nor silently ignored: it is recorded as REQUIRED BUT UNMAPPED (Section 5) and surfaced to HR/engineering for classification before the next upload, without blocking the current one.

8.2 SheetJS vs. alternatives, reassessed

Criterion	SheetJS (xlsx) — reading	ExcelJS — for the new HR review-workbook export (§9)
Fit for this workbook	Confirmed again on the current, wider (59-column) file, including reliably skipping the ~520 blank template rows and the TOTAL row — reads cached formula values correctly	n/a for reading; considered only for writing
Write-side styling/protection	SheetJS Community Edition does not support cell protection, locked formulas, or rich styling on write; that functionality sits behind the paid SheetJS Pro tier	ExcelJS supports cell protection, sheet protection, and styling on write in its free, MIT-licensed package — the capability the HR review workbook (Section 9) specifically needs
Ecosystem currency (2026)	Still the most widely used option (~7.8M weekly downloads); actively maintained	Still widely used for generation/streaming despite one third-party alpha project characterizing it as low-velocity; verify its release cadence at implementation time before committing
Recommendation	Keep for all ingestion (Billing Rate, any future machine-readable payroll file)	Adopt narrowly, only for generating the protected HR review workbook (Section 9) —

Criterion	SheetJS (xlsx) — reading	ExceUS — for the new HR review-workbook export (§9)
		not for ingestion, and not for the payslip PDF path (Section 11)

RECOMMENDATION: this is a change from v1, which saw no need for a second spreadsheet library. The new HR-editable review-workbook requirement (Section 9) specifically needs write-side cell protection that SheetJS Community Edition does not provide free of charge; introducing ExceUS narrowly for that one export path is a smaller cost than either paying for SheetJS Pro or hand-rolling protection with raw OOXML.

8.3 Storage and idempotency

Unchanged from v1: store the original workbook (object storage, content-hash keyed) and normalized, Staff-ID-keyed rows in PostgreSQL, linked by upload ID. New: the stored BillingRateUpload record captures accountingCompanyId and payrollPeriodId as first-class fields (Section 12), sourced from the Input tab, so multi-company processing (Section 13/21) does not need to re-derive company identity from row-level data alone.

9. Excel HR Review / Correction Architecture

New requirement (Section 5.8/10 of the brief): HR wants to review and tweak calculated payroll data in Excel before approval, not only in the dashboard. This is a controlled loop, not a second payroll engine.

Calculated Payroll (backend, authoritative)

↓

System-generated HR Review Workbook (ExceUS, protected)

↓

HR edits permitted fields only

↓

HR uploads the edited workbook

↓

System validates the workbook (version, period, company, Staff IDs, protected-cell integrity)

↓

System diffs original vs. edited values, field by field

↓

System recalculates only the affected employees

↓

HR reviews the diff and the recalculated results

↓

HR approves

↓

Immutable payslip version created

9.1 Workbook controls

- Hidden metadata embedded in the workbook (a hidden sheet or defined names): reviewWorkbookVersionId, payrollPeriodId, accountingCompanyId, sourceBillingRateUploadId, calculationEngineVersion, and an export hash — checked on re-upload before anything else runs.

- Cell/sheet protection (ExcelUS) locks every system-calculated cell (Consultant Gross Pay, Gross Earnings, WHT, Total Deductions, Net Service Fee, and all Billing-Rate-sourced input values); only the designated HR-editable cells (Other Deduction, and any explicitly approved correction fields) are left unlocked.
- A stale-workbook check: if the underlying Billing Rate has been re-uploaded (new file hash) since this review workbook was exported, the re-upload is rejected with a clear message rather than silently recalculating against a moved target.
- A malformed/malicious workbook is handled with the same file-type, size, and structural checks as the primary Billing Rate importer (Section 8) — the review workbook is not a trusted, exempt input merely because HR produced it.
- Duplicate submission of the same review workbook (same export hash) is detected and rejected as a no-op, not reprocessed.

9.2 Field editability

Read-only / system authoritative	HR-editable / controlled override
Staff ID; every Billing-Rate-sourced value (Base Fee, Days, all payslip-mapped allowances and their arrears); Consultant Gross Pay; Gross Earnings; WHT; Total Deductions; Net Service Fee; calculation engine version	Other Deduction (post-reconciliation, Section 6.4); any single field the current Allowance Definition configuration explicitly flags as HR-overridable for a documented business reason

This split is derived from the business process, not chosen arbitrarily: everything the Billing Rate already states is source data HR corrects by re-uploading a corrected Billing Rate (Section 3, "Billing Rate is the source of truth"), not by editing the review workbook; the review workbook exists specifically for the one category of value — Other Deductions — that the process defines as HR-supplied after the fact.

10. Dynamic Payslip / Arrears Architecture (mandatory)

FACT (observed in current artifacts): the current Billing Rate already implements a partial version of this idea structurally: nine allowances (Accommodation, Time Off Transport, Transport, Feeding, OutStation, Public Holiday, Weekend Work, Overtime, Present On Time Off) each have an adjacent "ARREARS —" companion column. The remaining payslip-mapped allowances (End of Contract Bonus, Joining Bonus, Leave Encashment, Rig, Job Bonus, Performance Bonus, 13th Month, Christmas Bonus, Marriage Support Grant, Arrears Service Fees itself) currently have no arrears companion column at all.

This means the "dynamic" requirement is broader than what the Billing Rate template currently provisions for, and broader than the payslip template's own mechanism (two blank buffer rows, Section 2.5). A fixed-row payslip template — with or without a couple of spare rows — is not sufficient once the number of allowance/arrears lines an employee can have is genuinely variable.

10.1 Domain model

```
PayslipLine {
  id
  payslipCalculationId
  allowanceDefinitionId // FK into the classification config, Section 5.1
  kind: "BASE" | "ARREARS" // an arrears line always references its parent BASE line
  parentLineId? // set only when kind = "ARREARS"
  amount: Decimal
  taxClass // inherited from the parent allowance's current classification
  sortOrder // for predictable, stable rendering
  sourceCell? // provenance, e.g. "ALLOWANCE SPREAD SHEET!AA16" for an arrears value}
```

10.2 Design answers to the brief's specific questions

- Separate calculation components: yes — an arrears line is its own PayslipLine row, summed alongside its parent in Gross Earnings/Net Service Fee, never merged into the parent's amount. This is what makes double counting structurally impossible: the sum is over line rows, and each Billing Rate column (base or arrears) produces exactly one line row.
- Tax status: inherited from the parent allowance's current Allowance Definition classification at calculation time, not re-decided per arrears line.
- Storage: arrears are stored as their own rows (never netted into the current-period allowance value), preserving the exact amounts for audit even if a later correction changes only one of the two.
- Zero/blank handling: a zero or blank arrears cell produces no PayslipLine row at all (consistent with "empty optional cell = 0" from v1) — the dashboard and PDF simply show nothing extra for that allowance, rather than a visible zero line.
- Multiple arrears entries: the model supports more than one arrears line per allowance per period if a future Billing Rate template adds them (e.g. "Arrears – Month 1", "Arrears – Month 2"); nothing in the schema caps the count.
- Auditability: every line's sourceCell and the calculation trace (Section 19) together show exactly which Billing Rate cell produced which amount, base or arrears, indefinitely.
- Excel review compatibility: the HR review workbook (Section 9) renders each PayslipLine as its own row so HR sees exactly what the engine calculated, in the same shape as the Billing Rate's own base/arrears column pairs.

10.3 Consequence for payslip generation

Because the number of PayslipLine rows varies per employee, the payslip cannot remain a fixed-address Excel template edited cell-by-cell (v1's original recommendation) without either truncating extra lines or leaving unpredictable blank rows. This is resolved in Section 11.

11. Payslip Generation Technology Decision

Option	Dynamic line items	Preserves current visual design	Verdict
A. Populate the existing Excel template → PDF (v1's original lean)	Poor — requires runtime row insertion into a formatted, formula-bearing template, which risks breaking merged cells, print areas, and pagination each time the line count changes	Highest, if row insertion works cleanly, but that reliability degrades exactly as line count grows — the scenario this section exists for	Not recommended as the sole path now that dynamic lines are mandatory
B. HTML/CSS template → PDF (headless Chromium)	Excellent — a templating engine naturally repeats a "line item" block for however many PayslipLine rows exist, with ordinary CSS pagination	High, achievable by matching the existing template's fonts, spacing, and column layout in CSS; requires one careful design pass to match the current look, then stays stable	Recommended production approach
C. Dedicated PDF/reporting engine	Good, if the library supports repeating sections	Medium — typically less pixel-fidelity control than HTML/CSS for matching an existing, specific design	Not adopted — no concrete advantage over B for this template design

RECOMMENDATION: move payslip generation to an HTML/CSS template rendered to PDF via headless Chromium (Puppeteer or Playwright), reserving the existing Excel template for two narrower, still-valuable roles: (1) the HR review workbook export (Section 9), where Excel's native protection/editability is the actual requirement, and (2) an optional

"raw data" export for HR's own records. The HTML template is built once from the current payslip's exact header block, fonts, and two-column earnings/deductions layout, with a repeating line-item partial that renders one row per PayslipLine (Section 10) in sortOrder, indenting an ARREARS line under its BASE line. The final PDF represents exactly the approved PayslipCalculation's line items — no template row is ever left blank or truncated.

12. Database / Prisma Design

Extends v1's schema with the multi-company/period model, the dynamic line-item model (Section 10), and the review-workbook loop (Section 9). Entity names are chosen for this business process, not copied from the brief's example list.

```
model AccountingCompany {
  id String @id @default(uuid())
  name String @unique // e.g. "ODUM ENERGY FZE - TRAIN A"
  uploads BillingRateUpload[]
}

model PayrollPeriod { // the parent period an accounting company's upload belongs to
  id String @id @default(uuid())
  periodStart DateTime
  periodEnd DateTime
  month String
  year Int
  uploads BillingRateUpload[]
  @@unique([periodStart, periodEnd])
}

model BillingRateUpload {
  id String @id @default(uuid())
  fileHash String @unique
  objectKey String
  accountingCompanyId String
  accountingCompany AccountingCompany @relation(fields: [accountingCompanyId], references: [id])
  payrollPeriodId String
  payrollPeriod PayrollPeriod @relation(fields: [payrollPeriodId], references: [id])
  status BillingRateUploadStatus @default(UPLOADED)
  employees BillingRateEmployee[]
  @@unique([accountingCompanyId, payrollPeriodId]) // one upload per company per period
}

model BillingRateEmployee { /* as in v1, plus accountingCompanyId for cross-company duplicate checks */ }

model AllowanceDefinition { // Section 5.1 configuration table
  id String @id @default(uuid())
  canonicalName String @unique
  sourceSheet String
  sourceHeader String
  classification AllowanceClass // PAYSIP_MAPPED_TAXABLE / PAYSIP_MAPPED_NONTAXABLE / IGNORED / MANUAL / UNMAPPED
  affectsGrossEarnings Boolean
  affectsNetServiceFee Boolean
  supportsArrears Boolean
  displayLabel String
  sortOrder Int
  effectiveFrom DateTime @default(now())
}

model Payslip { /* as in v1: stable per-employee-per-period identity, now also scoped by accountingCompanyId */ }
model PayslipVersion { /* as in v1: append-only, immutable once APPROVED */ }
```

```

model PayslipCalculation { /* as in v1, plus engineVersion */
  lines PayslipLine[]      // Section 10.1
}

model PayslipLine {          // Section 10.1, persisted verbatim
  id          String @id @default(uuid())
  payslipCalculationId String
  allowanceDefinitionId String
  kind        PayslipLineKind // BASE | ARREARS
  parentLineId String?
  amount      Decimal @db.Decimal(18,4)
  taxClass    AllowanceClass
  sortOrder   Int
  sourceCell   String?
}

model ReviewWorkbook {      // Section 9
  id          String @id @default(uuid())
  payslipCalculationId String @unique
  exportHash   String @unique
  exportedAt   DateTime @default(now())
  reuploadedAt DateTime?
  diff         Json?        // field-by-field original-vs-edited, once re-uploaded
}

// PayslipApproval, PayslipDelivery, AuditLog: unchanged in shape from v1, now additionally
// scoped by accountingCompanyId/payrollPeriodId for company-level reporting (Section 21).

```

RECOMMENDATION: one PayrollPeriod can contain up to four BillingRateUpload rows (one per accounting company), each processed and approved independently — a company’s upload failing validation must never block another company’s payslips. A period-level dashboard view aggregates status across the (up to four) uploads for HR’s overall visibility without merging their underlying data.

13. State Machine

Extends v1’s state machine with the review-workbook loop (Section 9) inserted between CALCULATED and APPROVED.

State	Entered when	Valid next states
UPLOADED	Billing Rate accepted for one accounting company/period	VALIDATING
VALIDATING → VALIDATED / VALIDATION_FAILED	Structural + row validation (Section 8, 16)	VALIDATED, or terminal VALIDATION_FAILED
CALCULATING → CALCULATED	Calculation engine runs (Section 6, 10)	GENERATED
GENERATED	PDF produced (Section 11) and, optionally, a review workbook exported (Section 9)	UNDER_REVIEW
UNDER_REVIEW	HR opens the payslip/batch, or has an outstanding review workbook out for edits	APPROVED, CORRECTIONS_REQUIRED
CORRECTIONS_REQUIRED	HR rejects, or re-uploads an edited review workbook that changes a value	CALCULATING (re-enters as a new version)

State	Entered when	Valid next states
APPROVED	HR approves — immutable from this point	DELIVERY_PENDING
DELIVERY_PENDING → DELIVERED / DELIVERY_FAILED (per channel)	Independent per email/WhatsApp, as in v1	DELIVERY_FAILED retries to DELIVERY_PENDING

14. HR Dashboard Workflow

Select Accounting Company + Payroll Period

↓

Upload that company's Billing Rate

↓

Validation Results (per-row pass/fail)

↓

Calculation Summary

↓

Optional: Export HR Review Workbook → edit → re-upload → view diff → recalculate

↓

Review Payslips (PDF preview, per-line calculation trace)

↓

Approve (bulk or per-employee)

↓

Optional Email Send

A period-level view rolls up all accounting companies' upload statuses for the same PayrollPeriod (Section 12) without merging their data, and a "Find Employee" screen (Section 15) works by Staff ID across whichever company that Staff ID belongs to.

15. Single-Employee Correction Workflow

Find Staff ID

↓

Select Period (and accounting company, if the Staff ID appears in more than one)

↓

View current/previous PayslipVersions

↓

Correct permitted inputs (Other Deduction, or re-point at a corrected Billing Rate row)

↓

Recalculate

↓

Diff against the previous version

↓

Review

↓

Approve



Optional re-send

Unchanged in principle from v1: the previous approved version is never overwritten; delivery history stays version-specific; the correction reason is mandatory and audited.

16. Email Distribution

Unchanged from v1 in design, resized for scale: queue-backed batch sending (BullMQ/Redis), retries with backoff, duplicate-send prevention per (payslipVersionId, channel), provider delivery-status webhooks (Resend, reusing the existing MailerService), and verified-address-only delivery. At an evidenced ~76 payslips for one company (and up to ~570 across all four in a period), the send should still be chunked and rate-limited against the provider's per-minute sending limits rather than enqueued as one unbounded batch — confirm current limits against the provider's live documentation at implementation time, since these change independently of this research.

17. WhatsApp Employee Retrieval

SUPPLIED BUSINESS RULE: confirmed flow: employee requests payslip → system identifies the WhatsApp number → asks for Staff ID → matches Staff ID to that same, already-identified employee → checks for an approved payslip → delivers the PDF.

- Identity binding: the inbound MSISDN resolves to exactly one Employee first (as in v1); Staff ID is then requested and checked only against that already-resolved employee — never used as a standalone lookup across the whole employee base. A correct Staff ID typed by someone who is not that employee, from that employee's own phone, still fails if the phone is not on file for that Staff ID.
- Compensating controls, since Staff ID is a modest secret (colleagues, HR, and printed documents can expose it): rate-limit payslip-request attempts per phone number and per Staff ID; escalate to a human HR officer after a small number of failed matches within a short window; log every attempt (success or failure) as a security-relevant audit event.
- Edge cases the brief specifically calls out: an employee with no registered phone, or a phone number shared/changed — both route to human HR escalation rather than a fallback self-service path, since a fallback would reintroduce exactly the risk the Staff-ID check exists to close.
- No approved payslip available — respond plainly and offer escalation; never expose a draft, rejected, or superseded version, and never reveal that a payslip exists in a not-yet-approved state.
- Multiple periods/history: the WhatsApp flow only ever serves the current, latest APPROVED version for the employee; historical retrieval (if ever offered) would be a deliberate, separate feature, not a side effect of this flow.

RECOMMENDATION: Staff ID plus phone possession is an adequate, proportionate control for this MVP given HR's explicit choice, provided the rate-limiting and escalation controls above are implemented alongside it — not deferred. This is flagged, not silently accepted, because payroll data is more sensitive than the typical WhatsApp self-service use case; if incidents occur post-launch, the next reasonable step is a one-time PIN sent to the employee's registered email as a third factor.

18. Security and Privacy

Engineering controls (RBAC/least-privilege for the HR payroll role, IDOR prevention on every payslip/employee lookup, encrypted object storage, short-lived signed URLs, no public payslip links, encryption in transit and at rest, secret

management, and file-upload hardening against malformed/malicious workbooks) are unchanged from v1 and remain in force.

18.1 Nigerian regulatory context (current, at time of writing)

FACT (observed in current artifacts): the Nigeria Data Protection Act 2023 (NDPA) is Nigeria's first comprehensive statutory data-protection law, signed 12 June 2023, and came into full effect from September 2025, superseding the earlier NDPR (2019). The Nigeria Data Protection Commission (NDPC) is the independent regulator; its 2025 General Application and Implementation Directive (GAID) is the current implementing guidance. This is engineering-relevant factual context, not legal advice — confirm current obligations with InterTech's counsel before go-live.

- Payslip data (compensation, bank account numbers, Staff IDs, phone numbers) is personal data under the NDPA; the module's existing audit-log-retention posture (archive before deletion, never permanent destruction on a routine schedule) aligns with the Act's accountability principle and should be extended to payslip/PDF retention, not treated as a separate policy.
- Least-privilege access to compensation data, encryption at rest and in transit, and an incident-response process are all consistent with the NDPA's security-of-processing expectations — the controls already planned in Section 18 (engineering) and v1's equivalent section satisfy this at a design level; a documented Data Protection Impact Assessment for this module is a reasonable, low-cost step given the sensitivity of the data and is recommended as a pre-launch action.
- Spreadsheet formula-injection risk (a malicious formula in a re-uploaded review workbook, Section 9) should be mitigated at the ingestion layer — sanitize or strip formulas from any cell HR is not permitted to control, and never evaluate a formula from an untrusted upload.

19. Auditability and Traceability

Every PayslipLine (Section 10.1) carries its own sourceCell, so tracing a final PDF figure back to source is now line-item-level, not just calculation-level: for any amount on an approved payslip, the audit trail shows the exact Billing Rate cell (or "HR-entered" for Other Deduction) that produced it, the AllowanceDefinition version in force at calculation time, the calculation engine version, who approved it and when, and which version superseded which. This directly answers every question listed in the brief's Auditability section (who uploaded, which company, which period, which rows failed, what HR changed via the review workbook, who approved, which version was delivered through which channel).

20. Idempotency / Duplicate Prevention

- File-level: fileHash unique per BillingRateUpload, as in v1.
- Company+period level: @@unique([accountingCompanyId, payrollPeriodId]) prevents the same company's Billing Rate being processed twice for the same period; a genuine correction is a new upload (new hash) still bound to the same company/period, producing a new PayslipVersion per affected employee (Section 15), not a duplicate Payslip.
- Review-workbook level: exportHash unique; a re-upload of the same export is a detected no-op (Section 9.1).
- Delivery level: unique, non-failed PayslipDelivery per (payslipVersionId, channel) prevents duplicate email/WhatsApp sends, as in v1.
- Cross-company duplicate Staff ID: a Staff ID appearing in two different accounting companies' uploads for the same period is not automatically an error — an employee could plausibly be billed through more than one company — but it is flagged for HR visibility rather than silently allowed to produce two unrelated payslips without HR being aware of the overlap.

21. Performance for ~570 Employees / Four Accounting Companies (Re-Sized)

FACT (observed in current artifacts): once blank template rows and the TOTAL row are correctly excluded (§8.1), the single inspected Odum workbook has 76 real employees, not 595. HR's ~570-across-four-companies estimate is therefore consistent with the evidence and is used directly below, with headroom rather than a hard ceiling, since individual companies will vary in size.

- Ingestion and validation for a company's upload (observed at 76 real employees, but sized with headroom to ~200 per company) should still run as a background job, not inline in the HTTP request; the controller returns immediately with the upload ID and the dashboard subscribes to progress via the existing Socket.IO realtime channel, exactly as v1 recommended for the smaller 74-row case.
- Calculation and PDF generation are queued per employee (BullMQ), allowing four companies' uploads to process concurrently without one company's batch starving another's queue — per-company queue priority or separate queues is worth evaluating once real concurrency numbers are known.
- Email sending for up to ~570 payslips across a period (all four companies combined) should be chunked against the provider's current rate limits (verify live at implementation time) rather than enqueued as one unbounded batch.
- Object storage is organized by accountingCompanyId/payrollPeriodId/employeeId prefixes so per-company retrieval and any future per-company access scoping stay efficient without a schema change.
- Database indexes: (staffId), (accountingCompanyId, payrollPeriodId), and (payslipId, versionNumber) cover the dashboard's primary lookup patterns at this scale without additional tuning; re-evaluate only if real query latency data says otherwise.

22. Testing Strategy

Layer	Coverage (new/updated for v2)
Calculation	All items in Section 6, including the six-add-term Net Service Fee formula, dynamic arrears (no double counting), Performance Bonus/13th Month as taxable, Christmas Bonus/Marriage Support Grant as non-taxable, precision (no rounding)
Excel ingestion	Header-alias resolution against the current 59-column workbook; a deliberately moved column (simulating template drift) is caught, not silently followed; duplicate/missing Staff ID; malformed amounts
Review workbook	Stale workbook rejection; edited read-only cell rejection; duplicate re-upload; diff correctness; formula-injection stripped on ingest
Workflow	Four concurrent company uploads for one period; partial validation failure in one company not blocking another; single-employee correction after approval; re-send of a corrected version
Security	Wrong Staff ID against a correct phone (denied); correct Staff ID against the wrong phone (denied); repeated failed WhatsApp attempts trigger escalation and are audited; IDOR attempts against another employee's payslip ID
Regression	The current Odum workbook and the worked example in Section 7 become permanent fixtures, alongside the original Seun Odufejo files retained from v1
Load	A synthetic ~200-employee company upload (headroom above the observed 76) processed end-to-end within an agreed SLA, and four such uploads processed concurrently, before go-live

23. Error Handling / Exception Management

Class	Examples	Behaviour
Blocking	Structural drift; Total Days = 0; Days Worked > Total Days; missing/duplicate Staff ID; stale review workbook; edited read-only cell	Whole upload or whole correction attempt fails; nothing partial is committed
Warning	A newly seen, unclassified workbook column (REQUIRED BUT UNMAPPED); a Staff ID appearing in more than one company's upload for the same period	Upload proceeds; surfaced to HR/engineering for a decision, not silently ignored and not blocking
Informational	A blank optional allowance treated as 0; an arrears cell with no value producing no PayslipLine	No error, no warning — expected, normal data shape

24. Implementation Roadmap

Phase	Scope
0	Schema migration for the v2 model (AccountingCompany, PayrollPeriod, AllowanceDefinition, PayslipLine, ReviewWorkbook); seed AllowanceDefinition from Section 4/5
1	Header-driven Billing Rate importer + structural validation, tested against the current Odum workbook
2	Calculation engine with dynamic line items (Section 10); full unit suite against Section 7's worked example
3	HTML/CSS → PDF payslip generation (Section 11); dashboard review/approval
4	HR Excel review/correction workbook loop (Section 9)
5	Multi-company/period dashboard rollup; single-employee correction
6	Email delivery at up to ~570-employee, four-company scale
7	WhatsApp Staff-ID-verified retrieval, built and reviewed last given its security sensitivity
8	Load test at ~200 employees × 4 companies (headroom above the observed scale) before first live payroll cycle

25. Risks

Category	Risk
Technical	Template drift: the Billing Rate's column layout has already changed once between the two research passes; the header-driven mapping (Section 8) reduces but does not eliminate this risk if a header's wording itself changes
Payroll	The new LOAN/ADVANCE-DEDUCTIONS column (Section 3/26) could be misread by a future engineer as "the" Other Deduction source without the HR confirmation this report flags as outstanding
Operational	Four accounting companies uploading independently increases the chance of a missed or late upload for one company in a given period; the dashboard's period-level rollup (Section 14) should make a missing company's upload visually obvious
Security	Staff ID as a WhatsApp verification factor is not a strong secret (Section 17); the compensating controls specified there are necessary, not optional

Category	Risk
Compliance	NDPA obligations (Section 18) require InterTech legal sign-off on retention/DPIA before go-live; this report provides engineering-relevant context only
Business	If a future accounting company's workbook is not cleanly formatted (e.g. blank rows carry a stray non-zero value in only one of Staff ID/Personnel/Base Fee), the three-field row-validity rule (§8.1) could misclassify a real employee as a template row or vice versa — worth a one-time manual spot-check of each company's first live upload

26. Re-evaluation of Previous Risks / Unresolved Questions

Covered in full in Section 3's Risk/Resolution Table, per the brief's instruction that this section "use the exact items from Section 29" (i.e., what this document treats as the single consolidated table rather than repeating it).

27. Final Questions Still Requiring HR/Business Confirmation

Only genuinely open items — nothing HR already answered in the source-hierarchy Section 5 material is re-asked here.

Group	Question
Other Deductions	Is the new "LOAN / ADVANCE – DEDUCTIONS" column on the current Billing Rate meant to seed Other Deductions automatically in a future version, or is it unrelated to the payslip process? (Section 3/26)
Accounting-company structure	Should the ~520 blank template rows per company workbook be trimmed by HR before upload, or is the importer expected to always tolerate a large tail of blank rows reserved for future joiners? (Section 8.1)
Dynamic arrears	For the payslip-mapped allowances that currently have no arrears companion column in the Billing Rate (End of Contract Bonus, Joining Bonus, Leave Encashment, Rig, Job Bonus, Performance Bonus, 13th Month, Christmas Bonus, Marriage Support Grant), should the Billing Rate template be extended with arrears columns for these too, or are arrears only ever expected on the nine allowances that already have them? (Section 10)
Billing Rate structure	Should the four accounting companies eventually be required to submit Billing Rates using one single, centrally controlled template file, given how much the template has already changed once? (Section 8)
Approval	Should a correction that only changes Other Deduction (no change to any Billing-Rate-sourced figure) require the exact same full HR review as a correction that changes a calculated amount, or is a lighter-touch review acceptable for that one specific field? (Section 9/15)

28. MVP Scope

Must have

- Header-driven Billing Rate import + structural/row validation for a single accounting company's upload
- Deterministic calculation engine covering the full Section 6 formula set, including dynamic arrears (Section 10)
- PDF generation via the HTML/CSS approach (Section 11)
- HR review and approval with immutable, versioned approved payslips
- HR-triggered email delivery
- Staff-ID-verified WhatsApp retrieval with rate limiting and audit logging

- Full audit trail to the line-item level (Section 19)

Should have

- The Excel HR review/correction workbook loop (Section 9) — valuable but can follow the dashboard-only review path in a first release
- Period-level, multi-company dashboard rollup (Section 14)
- Single-employee correction workflow (Section 15) beyond the first full-batch cycle

Later / Phase 2

- Automated Loan/Advance-based Other Deduction sourcing, once Section 27's question is resolved
- A machine-readable payroll reconciliation import, if one ever becomes available (payroll remains human-only until then, Section 12 of the brief)
- A stronger WhatsApp verification factor (e.g. a one-time email PIN) if warranted post-launch

29. Final Recommended Architecture

One coherent production recommendation, consolidating this document:

Billing Rate (per accounting company, per period)

↓

Import (SheetJS, header-driven mapping)

↓

Validation (structural + row-level)

↓

Normalization (AllowanceDefinition-classified domain model)

↓

Calculation (deterministic TypeScript, dynamic PayslipLine model)

↓

Review Workbook (ExcelJS, protected) — optional loop — Diff — Recalculation

↓

HR Approval (immutable, versioned)

↓

PDF (HTML/CSS → headless Chromium)

↓

Email (HR-triggered, queued) / WhatsApp (employee-requested, Staff-ID-verified)

↓

Audit (line-item provenance, every state transition)

This keeps the platform's existing NestJS/PostgreSQL/Prisma/Next.js stack and existing spreadsheet-ingestion precedent, extends it with the configuration-driven allowance model and dynamic line-item model the current artifacts prove is necessary, and sizes the operational design to the workforce scale actually evidenced rather than the smaller estimate first given.

30. Technology Evaluation Summary

Concern	Recommendation	Basis
Excel parsing	SheetJS (xlsx)	Existing precedent; confirmed against the current, larger workbook (§8.2)
Excel generation (review workbook only)	ExceUS	Free, MIT cell/sheet protection SheetJS Community lacks (§8.2)
PDF generation (payslip)	Headless Chromium (Puppeteer/Playwright) rendering an HTML/CSS template	Only option that cleanly supports the mandatory dynamic line items (§11)
Queue processing	BullMQ over Redis	Consistent with v1; fits NestJS; handles per-company concurrent batches (§21)
Object storage	S3-compatible, company/period/employee-prefixed keys, short-lived signed URLs	Unchanged from v1, reorganized for multi-company scale
Email delivery	Existing Resend-based MailerService	Reuse existing platform integration; verify current rate limits at implementation time
WhatsApp document delivery	Existing WhatsappGraphClient (Meta Cloud API)	Already integrated in the platform
Observability	Structured logging + the existing audit module, extended with company/period scope	No new tool needed at this scale

31. Final Quality Rules — Self-Check

Rule	Satisfied by
Billing Rate is consistently the source of truth	Sections 3, 6.1, 9.2
Staff ID is the deterministic identity key	Sections 4, 17
WHT applied exactly once, only via Total Deductions	Section 6.3–6.4
Other Deductions remain a controlled HR/manual input for MVP	Section 6.4, flagged conflict noted in Section 3/26/27
Four standard non-taxable allowances represented	Sections 4, 6.5
Performance Bonus and 13th Month taxable	Sections 4, 6.2
Christmas Bonus and Marriage Support Grant non-taxable	Sections 4, 6.5
Extra Day Work source mapped to PRESENT ON TIME OFF	Section 4
Unused Billing Rate allowances not accidentally calculated	Section 5 classification model
Dynamic arrears without double counting	Section 10.2
No arbitrary rounding	Section 6, Decimal arithmetic throughout

Rule	Satisfied by
Approved versions immutable; corrections create new versions	Sections 9.1, 13, 15
HR approval mandatory; email HR-triggered	Sections 13, 16
WhatsApp retrieves only the employee's own approved payslip; Staff ID is a verification factor, not a lookup key	Section 17
Payroll reconciliation remains human-only	Sections 12 (brief), 25
Architecture supports ~570 across four companies	Section 21 (confirmed consistent with the corrected 76-employee Odum count, §2.1/8.1)
Current/updated artifacts used wherever available; missing evidence identified rather than guessed	Throughout — every FACT is cited to a specific file/cell; every gap is named in Section 27